THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — SCIENCE · MEDIUM LEVEL FULL-YEAR PAPER SET 2

Full Syllabus (Chapters 1–12) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. A solution is saturated at 30 °C. On heating to 50 °C it becomes: 1

(a) solid **(b)** gas **(c)** more saturated **(d)** unsaturated

2. A caterpillar changes to a butterfly through a stage called: 1

(a) egg **(b)** adult **(c)** pupa **(d)** seed

3. Which of the following can be reused? 1

(a) burnt coal **(b)** ash **(c)** glass bottles **(d)** smoke

4. A year on a planet is the time it takes to: 1

(a) cross another planet **(b)** rotate once

(c) complete one revolution around the Sun **(d)** rise

5. Why should a scientist note the unit with each measurement? 1

(a) The number alone has no meaning without it

(b) Units look nice

(c) Units make it larger

(d) Units are optional

6. Which is the correct order of size, smallest to largest? 1

(a) shrub, herb, tree **(b)** tree, herb, shrub

(c) tree, shrub, herb **(d)** herb, shrub, tree

7. Which part of a food label tells us what the food is made of? 1

(a) the colour of the packet **(b)** the shape of the packet

(c) the list of ingredients **(d)** the brand logo

8. In a toy, two magnets are placed with their like poles facing; the toy parts will: 1

(a) stick together **(b)** spin

(c) push away from each other **(d)** melt

9. The metal hook at the end of a measuring tape helps to: 1

(a) hold the tape against the edge of the object so that measuring starts from zero

(b) make the tape bend

(c) make the tape heavier

(d) show the time

10. Why are cooking utensils such as pans often coated with a non-stick layer? 1

(a) to make them dissolve **(b)** to make them heavier
(c) to make them transparent **(d)** food does not stick to the surface

11. Temperature of a cup of tea is 80 °C, the room is 30 °C. After a long time, the tea's temperature will be about: 1

(a) 30 °C **(b)** 80 °C **(c)** 50 °C **(d)** 100 °C

12. A farmer in a dry region wants to conserve water for irrigation. A good method is: 1

(a) flooding the field **(b)** leaving taps open
(c) cutting trees **(d)** drip irrigation

13. Why do we heat the salt solution slowly towards the end during evaporation? 1

(a) to cool **(b)** to dissolve more
(c) to avoid spurting **(d)** to freeze

14. A flower produces seeds. Seeds on sowing give plants. This shows: 1

(a) digestion **(b)** respiration **(c)** reproduction **(d)** excretion

15. In 100 L of air, about how much oxygen is there? 1

(a) 50 L **(b)** 21 L **(c)** 78 L **(d)** 1 L

16. Why do we see only one side of the Moon from Earth? 1

(a) it rotates once in the time it orbits the Earth
(b) Earth blocks it
(c) it never rotates at all
(d) the other side is dark

17. To test whether seeds need water to germinate, which variable should be changed? 1

(a) the day of the week
(b) the type of seed, water and light all together
(c) the presence of water
(d) the colour of the pot

18. A plant has a thin soft stem, a few leaves, and is about 30 cm tall. It is probably a: 1

(a) herb **(b)** large climber **(c)** tree **(d)** shrub

19. Which combination gives a more balanced meal? 1

(a) roti, dal, vegetables and curd **(b)** only chips
(c) only sweets **(d)** only rice

20. A rod repels the north pole of a magnet. The end of the rod near the north pole is: 1

(a) non-magnetic **(b)** south pole **(c)** north pole **(d)** neutral

SECTION — B (Very Short Answer Type) 6 × 2 = 12

Answer in about 20–30 words each.

21. A clinical thermometer reading is 39 °C. Interpret it. 2

22. Why does the temperature stay constant at 100 °C when water boils continuously? 2

23. Why is evaporation unsuitable to get sand from a sand-water mixture? 2

24. Why is a dry seed considered living although it shows no activity? 2

25. Why is soil called a 'living' resource? 2

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26. Why is a day on Earth different from a year on Earth? 2

SECTION — C (Short Answer Type) 6 × 3 = 18

Answer in about 40–60 words each.

27. A mixture contains wheat, stones, iron nails and sand. Suggest a plan to separate. 3

28. Explain why fire is not living though it grows, needs air and gives out waste. 3

29. Describe an experiment to show that air is needed for burning. 3

30. Explain, with reasons, why life is possible on Earth but not on the Moon. 3

31. Describe how a fair test can be set up to find which paper towel absorbs the most water. 3

OR

What are the adaptations of fish for survival? Give three. 3

32. A person eats a balanced diet but never drinks water. What would happen? 3

OR

How can the poles of an unmarked bar magnet be identified with another marked magnet? 3

SECTION — D (Long Answer Type) 6 × 5 = 30

Answer in about 100–150 words each.

33. Plan an experiment to show that plants respond to light and describe expected observations. 5

34. Explain, with the help of an experiment, how to compare the water-holding capacity of sand, clay and loam. 5

35. Explain how to find the north direction at night using Saptarishi and Pole star, and also use the Moon's phases to estimate the date in the lunar month. 5

36. Rahul measured the length of his desk four times and got 120 cm, 121 cm, 119 cm and 120 cm. What is the best value? Why repeat? 5

OR

Why do some animals live in groups and others alone? Give examples and reasons. 5

37. Describe how cooking affects nutrients, with examples. 5

OR

Compare the behaviour of a magnet and an iron piece when brought near a bar magnet. 5

38. Explain with examples the concept of relative motion for 5
(a) a train passenger and
(b) the Sun as seen from Earth.

OR

Describe how to make a simple filter to separate a mixture of sand and salt in water. 5

END OF QUESTION PAPER Total: 80 Marks

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